



Lecture 1 — After class

Introduction to Quantitative Biology | The simplest model of gene expression, and noisy divisions

This sheet contains everything we derived in class, step by step, so that you can redo it alone. Read it with the board photos or your notes next to it. The exercises at the end use only what is here. Exercise 5 is marked **ADVANCED**.

The tags [Python notebook](#) [Section 2b](#) point to the sections of the class notebook (IntroQBio_L1_notebook.ipynb) where the same result is plotted.

1. What is quantitative biology?

In the last decades biology has become very good at making lists of parts: genomes, the set of RNAs in a cell, the set of proteins. A list of parts, however, does not tell us how a cell grows, how it senses its environment, or how it decides what to do. For that we need to understand how the parts interact, and how fast, and how much.

Quantitative biology is **not** biology plus numbers, and it is **not** biology plus computational tools. It uses numbers, models and tools to make **predictions** and to obtain biological **understanding**. Models are useful for three reasons:

- a model forces us to propose a *mechanism* that explains the data;
- a model makes *predictions* that an experiment can test — and when the prediction fails, we learn which assumption was wrong;
- a model *simplifies*: it leaves out details on purpose, so that one aspect of reality becomes visible.

KEY TAKEAWAY

The modelling loop, which we follow in every lecture: (1) ask a biological question → (2) build a model and state its assumptions → (3) solve it, choosing the right tool → (4) interpret the result and compare it with numbers → (5) refine the model.

Some numbers of an *E. coli* cell (Alon, Tables 1.1 and 1.2)

About 4×10^6 proteins of about 4500 types, so on average about 900 copies of each type (many proteins, for example transcription factors, have far fewer). One protein per cell corresponds to a concentration of about 1 nM. Making a protein takes about a minute; a typical protein lives about an hour; a generation lasts from 20 minutes to several hours. These timescales differ by orders of magnitude, which is what allows us to neglect the fastest steps in the model below.

2. The simplest model of gene expression

The model and its assumptions

A gene is switched on. Its protein has concentration $Y(t)$. The protein is **produced** at a constant rate β and **removed** at a rate αY : each protein has the same probability per unit time α of being removed, so

the more proteins there are, the faster they are removed. Removal has two parts: active degradation, and dilution because the cell grows. Following Alon [1]:

$$\frac{dY}{dt} = \beta - \alpha Y, \quad \alpha = \alpha_{\text{dil}} + \alpha_{\text{deg}}.$$

β is a concentration per unit time; α is in units of 1/time.

Assumptions we made

- The cell is well mixed: one number Y describes it.
- β and α are constant in time, and production does not depend on Y .
- The minutes needed for transcription and translation are neglected: they are short compared with the time for the protein level to change (about one hour).
- Y is a continuous, deterministic quantity. Sections 3 and 4 show where this stops being enough.

Reading the equation

If $Y < \beta/\alpha$, then $dY/dt > 0$ and Y increases. If $Y > \beta/\alpha$, it decreases. At

$$Y_{\text{st}} = \frac{\beta}{\alpha}$$

production and removal balance and Y stops changing: this is the **steady state**. From both sides Y moves towards it, so it is stable.

Solving it: method 1, separation of variables

For any $dx/dt = f(x)$ with $x(0) = x_0$, separating the variables gives

$$\frac{dx}{f(x)} = dt \implies \int_{x_0}^{x(t)} \frac{dx'}{f(x')} = t.$$

This gives t as a function of x . To obtain $x(t)$ we must compute the integral and then invert the result. When one of the two steps is impossible, we solve the equation numerically (Lecture 2).

With $f(Y) = \beta - \alpha Y$, use $u = \beta - \alpha Y$, $du = -\alpha dY$:

$$\int_{Y_0}^Y \frac{dY'}{\beta - \alpha Y'} = -\frac{1}{\alpha} \ln \frac{\beta - \alpha Y}{\beta - \alpha Y_0} = t \implies \beta - \alpha Y = (\beta - \alpha Y_0) e^{-\alpha t}.$$

Dividing by α and rearranging gives the solution boxed below.

Solving it: method 2, homogeneous + particular solution

This method works for every **linear** ODE (an equation where Y and dY/dt appear only to the first power).

Step 1 — the homogeneous equation. Remove the term that does not contain Y : $dY/dt = -\alpha Y$. Its solutions are $Y_h(t) = C e^{-\alpha t}$, for any constant C (as in the sheet before class).

Step 2 — one particular solution. Find *any one* function that solves the full equation. The simplest guess is a constant, Y_p . Then $dY_p/dt = 0$, so $0 = \beta - \alpha Y_p$ and $Y_p = \beta/\alpha$: the steady state.

Step 3 — add them. $Y = Y_h + Y_p$ is a solution: its derivative is $-\alpha Y_h + 0$, and $\beta - \alpha(Y_h + Y_p) = -\alpha Y_h$ because $\beta - \alpha Y_p = 0$.

Step 4 — the initial condition. $Y(0) = C + \beta/\alpha = Y_0$ gives $C = Y_0 - \beta/\alpha$. Both methods give

$$Y(t) = \frac{\beta}{\alpha} + \left(Y_0 - \frac{\beta}{\alpha}\right) e^{-\alpha t}$$

Check: at $t = 0$ it gives Y_0 ; as $t \rightarrow \infty$ it gives β/α ; its derivative is $-\alpha(Y_0 - \beta/\alpha)e^{-\alpha t}$, which is equal to $\beta - \alpha Y(t)$.

[Python notebook](#) [Section 2a](#)

Response time

Start from $Y_0 = 0$: $Y(t) = Y_{\text{st}}(1 - e^{-\alpha t})$. The **response time** $T_{1/2}$ is the time to reach half of the steady state:

$$1 - e^{-\alpha T_{1/2}} = \frac{1}{2} \implies T_{1/2} = \frac{\ln 2}{\alpha}.$$

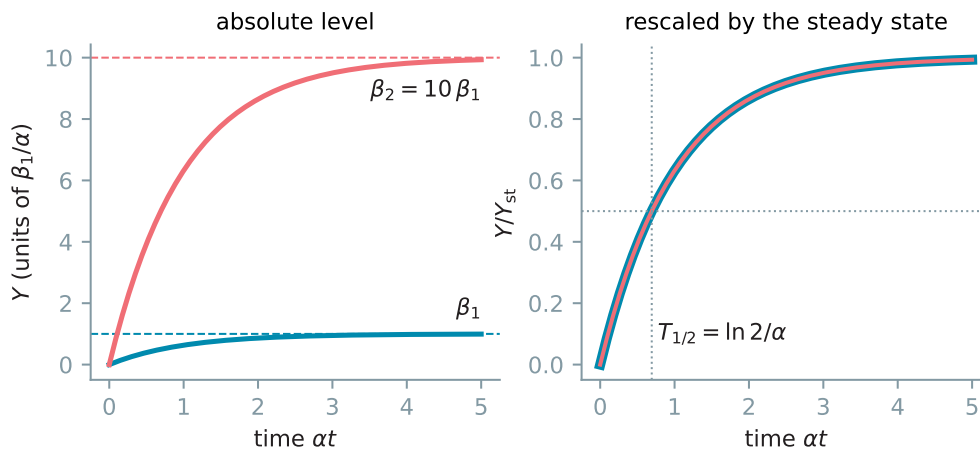


Figure 1: Two genes with the same α ; the second is produced ten times faster. Left: it reaches a ten times higher level. Right: divided by their own steady states, the two curves are the same, and both reach one half at $T_{1/2} = \ln 2/\alpha$. Time is in units of $1/\alpha$.

[Python notebook](#) [Section 2b](#)

KEY TAKEAWAY

The production rate β sets how much protein there is at steady state. Only the removal rate α sets how fast the steady state is reached.

Stable proteins respond in one generation

Many proteins are not actively degraded ($\alpha_{\text{deg}} = 0$), so only dilution removes them. If production stops, after one cell generation τ the volume has doubled and the concentration has halved. So $T_{1/2} = \tau$, and $\alpha_{\text{dil}} = \ln 2/\tau$ (Alon, Section 1.4.1). For an *E. coli* dividing every 30 min, $\alpha = 0.023 \text{ min}^{-1}$ and the response takes 30 min.

A cell can respond faster by degrading the protein actively (larger α). The price: to keep the same steady state β/α , it must produce more (β must grow with α).

[Python notebook](#) [Section 2c](#)

3. Switching off production

Question. A stable protein is at steady state. At $t = 0$ its gene is switched off. How many generations do we have to wait until the protein has disappeared?

With $\beta = 0$ the equation is the homogeneous one, $dY/dt = -\alpha Y$, and with $Y(0) = Y_{\text{st}}$:

$$Y(t) = Y_{\text{st}} e^{-\alpha t}.$$

For a stable protein $\alpha = \ln 2/\tau$. After n generations ($t = n\tau$), using $e^{-n \ln 2} = 2^{-n}$:

$$Y(n\tau) = \frac{Y_{\text{st}}}{2^n}.$$

The protein is halved at every generation. As time goes on, Y goes to zero, but it never becomes exactly zero in a finite time.

Counting molecules. Y is also a number of molecules per cell, and molecules come one by one. There is less than one molecule per cell when $Y_{\text{st}}/2^n < 1$, that is after

$$n = \log_2 Y_{\text{st}} \text{ generations.}$$

For $Y_{\text{st}} = 1000$ molecules this is about 10 generations (5 hours if $\tau = 30$ min); for 100 molecules 6.6 generations; for 10 molecules 3.3 generations.

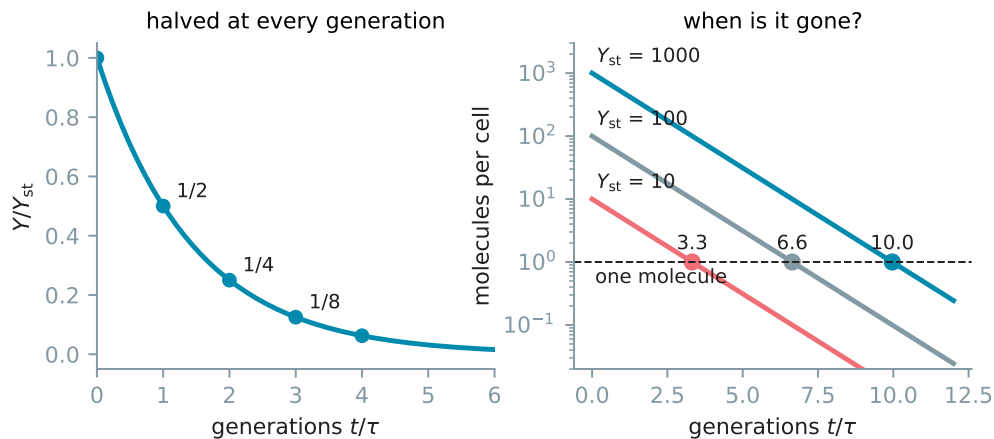


Figure 2: Production switched off at $t = 0$ for a stable protein. Left: the level is halved at every generation. Right: the number of molecules per cell, on a logarithmic axis, for $Y_{\text{st}} = 10, 100, 1000$. The dashed line is one molecule; it is reached after $\log_2 Y_{\text{st}}$ generations (3.3, 6.6 and 10.0).

[Python notebook](#) [Section 3](#)

Is it the same in every cell? The equation says that every cell has exactly $Y_{\text{st}}/2^n$ molecules after n generations. Halving, however, is what happens *on average*: at each division the molecules are shared at random between the two daughters. Section 4 computes how large the difference between two sister cells is.

4. Partitioning at cell division

When a cell divides, how are its N molecules shared between the two daughters? This section follows Phillips *et al.* [2], Chapter 2, pp. 43–48.

Assumptions

The two daughters have the same volume. Each molecule ends up in daughter 1 with probability $1/2$, independently of the others. No molecule is actively sent to one side.

Each molecule flips a coin

With these assumptions, sharing N molecules is exactly tossing N fair coins and counting the heads. In the bootcamp, the probability of k heads in N tosses, each with probability p of heads, was the **binomial distribution**

$$P(k) = \binom{N}{k} p^k (1-p)^{N-k}, \quad \langle k \rangle = Np, \quad \text{Var}(k) = Np(1-p).$$

Here the “heads” are the molecules that go to daughter 1: $k = n_1$ and $p = 1/2$. Since $p^{n_1}(1-p)^{N-n_1} = (1/2)^{n_1}(1/2)^{N-n_1} = 1/2^N$, daughter 1 receives n_1 molecules (and daughter 2 receives $n_2 = N - n_1$) with probability

$$P(n_1) = \binom{N}{n_1} \frac{1}{2^N}, \quad \langle n_1 \rangle = \frac{N}{2}, \quad \text{Var}(n_1) = \frac{N}{4}.$$

The standard deviation is $\sigma_{n_1} = \sqrt{N}/2$, so the **relative noise** is

$$\frac{\sigma_{n_1}}{\langle n_1 \rangle} = \frac{\sqrt{N}/2}{N/2} = \frac{1}{\sqrt{N}}.$$

It is 0.32 for $N = 10$, 0.10 for $N = 100$ and 0.03 for $N = 1000$. For $N = 10$, daughter 1 receives exactly 5 molecules with probability $252/1024 \approx 0.25$ only.

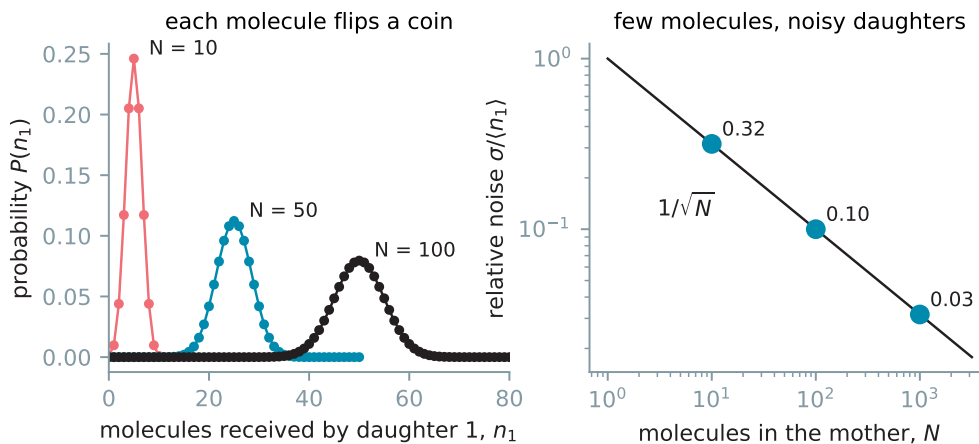


Figure 3: Left: probability that daughter 1 receives n_1 molecules, for $N = 10, 50, 100$. Right: relative noise $1/\sqrt{N}$, with the values for $N = 10, 100$ and 1000 .

[Python notebook](#) [Section 4](#)

KEY TAKEAWAY

Two genetically identical daughters of the same mother can differ just because molecules are few and the sharing is random. The fewer the molecules, the larger the relative difference.

Counting molecules with a microscope

In an experiment we measure fluorescence, not molecules, and we do not know how bright one molecule is. Call ν the intensity of one molecule. Then

$$I_1 = \nu n_1, \quad I_2 = \nu(N - n_1), \quad I_{\text{tot}} = I_1 + I_2 = \nu N.$$

I_1 , I_2 and I_{tot} can be measured; N , n_1 and ν cannot. We compute the average of the squared difference between the two daughters.

Two rules for the variance

For a random number X and two constants a and c : $\text{Var}(aX) = a^2 \text{Var}(X)$, $\text{Var}(X + c) = \text{Var}(X)$. And if the mean of X is 0, then $\langle X^2 \rangle = \text{Var}(X)$.

Step 1. $I_1 - I_2 = \nu n_1 - \nu(N - n_1) = \nu(2n_1 - N)$.

Step 2. Its mean is $\nu(2\langle n_1 \rangle - N) = \nu(N - N) = 0$. So $\langle (I_1 - I_2)^2 \rangle = \text{Var}(I_1 - I_2)$.

Step 3. With the two rules ($a = 2\nu$, $c = -\nu N$): $\text{Var}(\nu(2n_1 - N)) = 4\nu^2 \text{Var}(n_1) = 4\nu^2 \cdot \frac{N}{4} = \nu^2 N$.

Step 4. Replace $N = I_{\text{tot}}/\nu$:

$$\langle (I_1 - I_2)^2 \rangle = \nu I_{\text{tot}}$$

What it is good for. Measure many divisions, and plot $\langle (I_1 - I_2)^2 \rangle$ against I_{tot} . The points should lie on a straight line through the origin, whose slope is ν : the brightness of one molecule. Once ν is known, the number of molecules in any cell is its intensity divided by ν . Rosenfeld *et al.* (*Science*, 2005) used this idea on real data (see the slides).

Exercises

Exercise 1: Changing the production rate

A gene is at steady state with production rate β_1 . At $t = 0$ the production rate suddenly becomes β_2 (removal rate α unchanged).

- What is $Y(0)$? Use the boxed solution of Section 2 to write $Y(t)$ for $t > 0$.
- Sketch $Y(t)$ for $\beta_2 > \beta_1$ and for $\beta_2 < \beta_1$. What is the case $\beta_2 = 0$?
- Show that the time to go halfway from the old to the new steady state is $\ln 2/\alpha$ in both cases.

You can compare your answer with Alon, Exercise 1.1.

Exercise 2: Putting numbers on it

An *E. coli* cell divides every 60 min. Protein A is stable. Protein B is also actively degraded, with a degradation half-life of 20 min (the time it would take to lose half of it by degradation alone).

- Compute α_{dil} and, for protein B, α_{deg} (in min^{-1}).
- Compute the response time $T_{1/2}$ of A and of B.
- A and B have the same steady-state concentration. How many times larger must β_B be than β_A ?
- Both proteins are at steady state with 1000 molecules per cell when their genes are switched off. After how many minutes is there less than one molecule per cell, for A and

for B? How many generations is that?

Exercise 3: One curve for all genes

Define a time without units $s = \alpha t$ and a level without units $y = Y/Y_{\text{st}}$.

- (a) Using $dY/dt = \alpha Y_{\text{st}} dy/ds$, show that the equation becomes $dy/ds = 1 - y$.
- (b) How many parameters are left? Use this to explain the right panel of Figure 1.

Exercise 4: Partitioning with numbers

- (a) A mother cell has $N = 100$ molecules. What are the mean and the standard deviation of n_1 ? Same question for $N = 10$.
- (b) For $N = 10$, what is the probability that daughter 1 receives no molecule? That one of the two daughters (either) receives none?
- (c) Redo the derivation of $\langle (I_1 - I_2)^2 \rangle$ without the variance rules: expand $(2n_1 - N)^2$, take the average of each term, and use $\langle n_1^2 \rangle = \text{Var}(n_1) + \langle n_1 \rangle^2$.
- (d) In an (invented) experiment, the points $\langle (I_1 - I_2)^2 \rangle$ against I_{tot} lie on a line of slope 250 (intensity units). A cell has intensity 5000. About how many molecules does it contain?

Exercise 5: ADVANCED — Unequal division

Suppose daughter 1 is larger, and each molecule goes to it with probability $p \neq 1/2$.

- (a) Give $\langle n_1 \rangle$ and $\text{Var}(n_1)$.
- (b) Compute the mean of $2n_1 - N$. Is it still 0?
- (c) Using $\langle X^2 \rangle = \text{Var}(X) + \langle X \rangle^2$, show that

$$\langle (I_1 - I_2)^2 \rangle = \nu^2 [N^2(2p - 1)^2 + 4Np(1 - p)].$$

- (d) Is $\langle (I_1 - I_2)^2 \rangle$ still proportional to I_{tot} ? What would you see on the plot if you assumed equal division but the cells divided unequally?

References

- [1] Uri Alon. *An Introduction to Systems Biology: Design Principles of Biological Circuits*. 2nd. CRC Press, 2019.
- [2] Rob Phillips et al. *Physical Biology of the Cell*. 2nd. Garland Science, 2013.